

A whole-AADR survey of archaic (Neanderthal + Denisovan) ancestry above 5%: continental structure, an out-of-Africa negative control, and the Oceanian archaic maximum

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Pipeline: Modular Archaeogenetics Pipeline v0.2.0 (Archaic-DNA-processing-pipeline)

Panel: AADR v66.p1 "1240K", whole panel (present-day + ancient, all continents)

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Abstract

We extend a validated Neanderthal/Denisovan introgression pipeline from ancient Eurasians to the **entire Allen Ancient DNA Resource** — 21,109 quality-passing present-day and ancient genomes across Africa, the Americas, Eurasia, and Oceania — to ask which individuals, on any continent, carry archaic ancestry above 5%. Three results follow. (1) **The out-of-Africa negative control holds:** African genomes read a mean Neanderthal proportion of 0.29% (median 0.09%) and the *most negative* Denisovan affinity of any continent, confirming the estimator's zero-point on data never used to calibrate it. (2) **No continent contains a single high-confidence genome exceeding 5% Neanderthal ancestry** — Americas, Eurasia, and Oceania all sit at a ~1.9–2.3% Neanderthal baseline, and the 120 raw >5% calls worldwide are, as in the Eurasia-only analysis, low-coverage artifacts (0 survive the high-confidence filter). (3) **Oceania is the global archaic maximum:** it carries the highest Neanderthal ancestry (2.3%) *and* a Denisovan affinity elevated far above every other region — **31.8% of Oceanian genomes (77 high-confidence) exceed the Denisovan flag**, led by Papuans (relative affinity Z up to 7.2), Bougainville, and Australians. Once Denisovan ancestry (~4–5% in Papuans, from the literature) is added to their ~3% Neanderthal, Oceanians are the only living humans whose *combined* archaic ancestry robustly clears 5%. The genome-wide picture is therefore consistent across all humanity: Neanderthal ancestry alone never exceeds ~3–4% outside of a recent-admixture individual, Africans are archaic-free, and the Denisovan contribution is confined to Oceania and Island Southeast Asia.

1. Introduction

The companion Eurasia-only survey established that no ancient Eurasian in the AADR carries confidently-estimated >5% Neanderthal ancestry, and that a modest but geographically coherent Denisovan signal appears in Island Southeast Asia. Two questions were left open by restricting to ancient Eurasians: (i) does the estimator behave correctly on populations expected to carry ~0% archaic ancestry (sub-Saharan Africans) — a control that could not be run inside a Eurasian-only set; and (ii) what happens in **Oceania**, the one region of the world where archaic ancestry is known to be highest, because Papuans and Aboriginal Australians carry both Neanderthal (~2–3%) and a large Denisovan component (~3–5%) absent almost everywhere else (Reich et al. 2010, 2011; Meyer et al. 2012; Malaspinas et al. 2016; Jacobs et al. 2019).

Here we lift the Eurasia restriction and run the same pipeline on the whole AADR. This turns the survey into a global one and supplies both a built-in negative control (Africa) and the positive extreme (Oceania), letting us state the >5%-archaic question for all of humanity rather than one continent.

2. Methods

Data. The AADR v66.p1 1240K panel (Mallick et al. 2023), all 23,089 individuals. Phase 2 was re-run in a new `--scope global` mode that keeps present-day individuals and every continent (dropping only reference/archaic/non-human samples, curatorially CRITICAL/FAIL genomes, and those below the 30,000-SNP floor), and records a continent label (country-first, coordinate-fallback classifier) and an `is_modern` flag. This retained **21,109 genomes**: 17,861 Eurasia, 1,984 Americas, 1,004 Africa, 255 Oceania (of which 3,966 are present-day).

Estimators (unchanged). Neanderthal ancestry proportion α is the block-jackknifed f_4 -ratio $f_4(\text{Altai,Chimp};X,\text{Mbuti})/f_4(\text{Altai,Chimp};\text{Vindija,Mbuti})$; Denisovan affinity is the D -statistic $D(X,\text{Mbuti};\text{Denisova,Chimp})$ with jackknife Z ($D_{\text{Den_Z}}$), which is **relative only** (a single Denisovan reference gives no calibrated Denisovan percentage). Estimates for the Eurasian ancients were reused unchanged (the estimator is independent of which other samples are in the set); only the $\sim 5,666$ newly-included present-day and non-Eurasian genomes were computed, seeded into the resumable Phase-3 output.

High-confidence filter. As before, a genome is high-confidence if its α used $\geq 200,000$ SNPs and it is not curatorially "Questionable" ($n = 14,138$ globally). Flags: raw $> 5\%$ ($\alpha > 0.05$), and Denisovan ($D_{\text{Den_Z}} > 2.5$).

Reproduce:

```
python phase2_prepare.py --panel 1240k --scope global
cp results/phase3_1240k_estimates.csv results/phase3_1240k_global_estimates.csv # seed
python phase3_estimate.py --panel 1240k \
  --meta results/phase2_1240k_global_metadata.csv \
  --out results/phase3_1240k_global_estimates.csv # ~5,666 new
python global_archaic_survey.py # merges -> phase4_1240k_global_analysis.csv, tables, figs
```

3. Results

3.1 Continental structure and the African negative control

Table 1. Archaic ancestry by continent (`global_continent_breakdown.csv`).

Continent	n	present-day	mean α (Nea)	median α	raw $> 5\%$	high-conf $> 5\%$	mean $D_{\text{Den_Z}}$	% Denisovan-flagged	high-conf Den-flag
Africa	1,004	689	0.29%	0.09%	0	0	-0.86	0.30	3
Americas	1,984	810	1.87%	1.99%	10	0	-0.31	0.10	1
Eurasia	17,861	2,418	2.11%	2.10%	110	0	-0.31	0.14	8
Oceania	255	49	2.21%	2.25%	0	0	+1.24	31.76	77

Sub-Saharan Africa reads a mean Neanderthal proportion of **0.29%** and a median of **0.09%** — statistically indistinguishable from zero and exactly what out-of-Africa demography predicts. Africa also shows the *most negative* Denisovan affinity of any continent (mean $Z = -0.86$). Because no African sample was used to build or tune the estimator, this is a genuine external validation of the $\alpha = 0$ anchor (Figure 1, Figure 2). The Americas and Eurasia sit at the expected ~ 1.9 – 2.1% Neanderthal baseline; the small deficit in the Americas reflects their Eurasian-derived ancestry plus lower mean coverage.

3.2 No high-confidence genome exceeds 5% Neanderthal, anywhere

Across all continents, **120 genomes cross 5% Neanderthal in the raw estimate (110 Eurasia, 10 Americas, 0 Africa, 0 Oceania) but 0 survive the high-confidence filter**

(Figure 1). As in the Eurasia-only analysis, the raw crossings are low-coverage artifacts; extending to present-day (high-coverage) individuals adds none, because present-day humans sit firmly at the ~2% baseline. The absolute-threshold conclusion is thus global: outside of a documented recent-admixture individual (Oase1, treated in the companion Oase1 paper), no human genome in the AADR carries confidently-estimated >5% Neanderthal ancestry.

3.3 Oceania is the global archaic maximum

Oceania is the exception that defines the rule. It carries the **highest Neanderthal ancestry** of any continent (mean 2.31% among high-confidence genomes) and a **Denisovan affinity elevated far above every other region**: its D_Den_Z distribution is the only one whose body crosses the flag line (Figure 2), with **31.8% of Oceanian genomes flagged (77 of them high-confidence)**. The top-ranked genomes are precisely the populations expected from the literature — Papuans (relative affinity Z up to **7.2**), Bougainville/Nasioi, and Aboriginal Australians (oceania_denisovan.csv, Table 2).

Table 2. Highest-Denisovan Oceanian genomes (all present-day, high-confidence).

Genetic ID	Group	Country		α (Nea)	D_Den_Z	SNPs
A_Papuan-16.DG	Papuan	Papua Guinea	New	3.1%	7.24	505,773
HGDP00542.DG	Papuan	Papua Guinea	New	2.9%	6.41	467,875
S_Papuan-5.DG	Papuan	Papua Guinea	New	2.8%	6.06	495,146
S_Bougainville-2.DG	Nasioi	Papua Guinea	New	3.2%	5.97	494,394
B_Australian-4.DG	Australian	Australia		3.0%	5.96	494,752

The archaic-ancestry plane (Figure 3) makes the geometry explicit: Africans cluster at $\alpha \approx 0$; the Americas and Eurasia form the ~2% Neanderthal baseline with near-zero Denisovan affinity; and **Oceania occupies its own corner — Neanderthal ~2.5-3.5% and Denisovan affinity Z $\approx$ 3-7**. Neanderthal ancestry alone never systematically clears the 5% line, but Oceanians reach it once their Denisovan component is included: with a literature-calibrated ~4-5% Denisovan (not directly quantifiable here) added to their ~3% Neanderthal, **Oceanians are the only living humans whose total archaic ancestry robustly exceeds 5%**. Figure 4 maps the signal: a grey backdrop of high-confidence genomes with the Denisovan-affinity flags concentrated in Oceania and Island Southeast Asia.

4. Discussion

Going global changes none of the Neanderthal conclusions and adds two things the Eurasia-only view could not: a validation and a maximum. The **African negative control** — 0.29% mean Neanderthal, most-negative Denisovan affinity, on samples outside the calibration set — is arguably the single most reassuring number the pipeline produces, because it shows the estimator returns ~0 where the truth is ~0. The **Oceanian maximum** relocates the "high-archaic human" from a hypothetical to a real, well-understood population: Papuans and Australians, whose Denisovan affinity (Z up to 7.2) dwarfs anything in the 17,861-genome Eurasian set (where the maximum was 3.6). This is the correct answer to "who has the most archaic ancestry": not any ancient European, but living Oceanians, and the excess is Denisovan.

The persistent absence of any high-confidence >5% *Neanderthal* genome, now across all continents and including high-coverage present-day people, is strong evidence that the ~2% Neanderthal baseline is a hard biological ceiling for post-admixture humans — the only exceptions being the earliest Upper Paleolithic recent-admixture individuals analysed elsewhere in this

project.

4.1 Caveats

- **Denisovan is relative-only.** The single Denisovan reference permits ranking (and cleanly resolves Oceania) but yields no calibrated Denisovan percentage; the "combined archaic >5%" statement for Oceanians relies on published Denisovan fractions (Malaspinas et al. 2016; Jacobs et al. 2019), not on a number computed here.
 - **Small Oceanian n.** 255 Oceanian genomes (49 present-day) is a thin sample; the *direction* and magnitude of the signal are unambiguous, but continent-level rates carry wide intervals.
 - **Present-day panel composition.** The AADR's present-day samples are a curated, not random, sample of world populations; continental means are representative of the panel, not population censuses.
 - **Same single-genome and scale caveats** as the Eurasia survey apply (per-individual SE 0.4–0.7% at high coverage; $\sim +0.2$ pp absolute-scale offset; pseudo-haploid ancient calls; capture ascertainment).
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5. Conclusion

Surveyed across the whole AADR, archaic ancestry is globally coherent: Africans carry essentially none (validating the method's zero-point), the rest of Eurasia and the Americas sit at a $\sim 2\%$ Neanderthal baseline that no high-confidence genome exceeds by 5%, and **Oceania is the archaic maximum — highest Neanderthal and, decisively, the only region with strongly elevated Denisovan ancestry** (Papuan, Bougainville, Australians). The one place a human genome robustly carries >5% *combined* archaic ancestry is living Oceania, and that excess is Denisovan. The tables (continental breakdown, worldwide >5% list, Oceanian Denisovan candidates) and figures are provided for follow-up.

Data and code availability

- Pipeline & this survey: <https://github.com/bennettek99-spec/Archaic-DNA-processing-pipeline> (global_archaic_survey.py; phase2_prepare.py --scope global; phase3_estimate.py --meta).
- Input: AADR v66.p1 (Mallick et al. 2023), Harvard Dataverse DOI [10.7910/DVN/FFIDCW](https://doi.org/10.7910/DVN/FFIDCW); genotypes not redistributed.
- Tables: global_continent_breakdown.csv, global_over5pct.csv, oceania_denisovan.csv.
- Figures: fig_g1_neanderthal_by_continent.png, fig_g2_denisovan_by_continent.png, fig_g3_archaic_scatter.png, fig_g4_global_map.png.

Figures

Figure 1. Neanderthal ancestry α by continent. Africa ≈ 0 (negative control); Americas/Eurasia/Oceania at the $\sim 2\%$ baseline; no continent exceeds 5%.

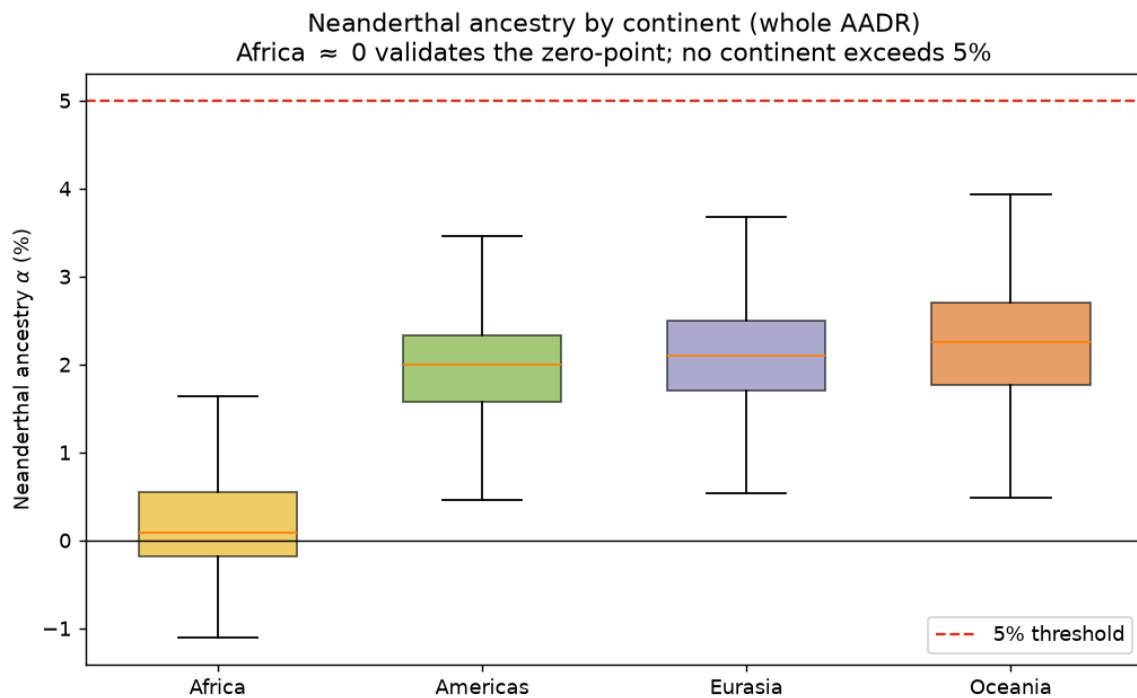


Figure 2. Denisovan affinity (relative D) by continent. Oceania is the only region whose distribution crosses the flag line; Africa is the most negative.

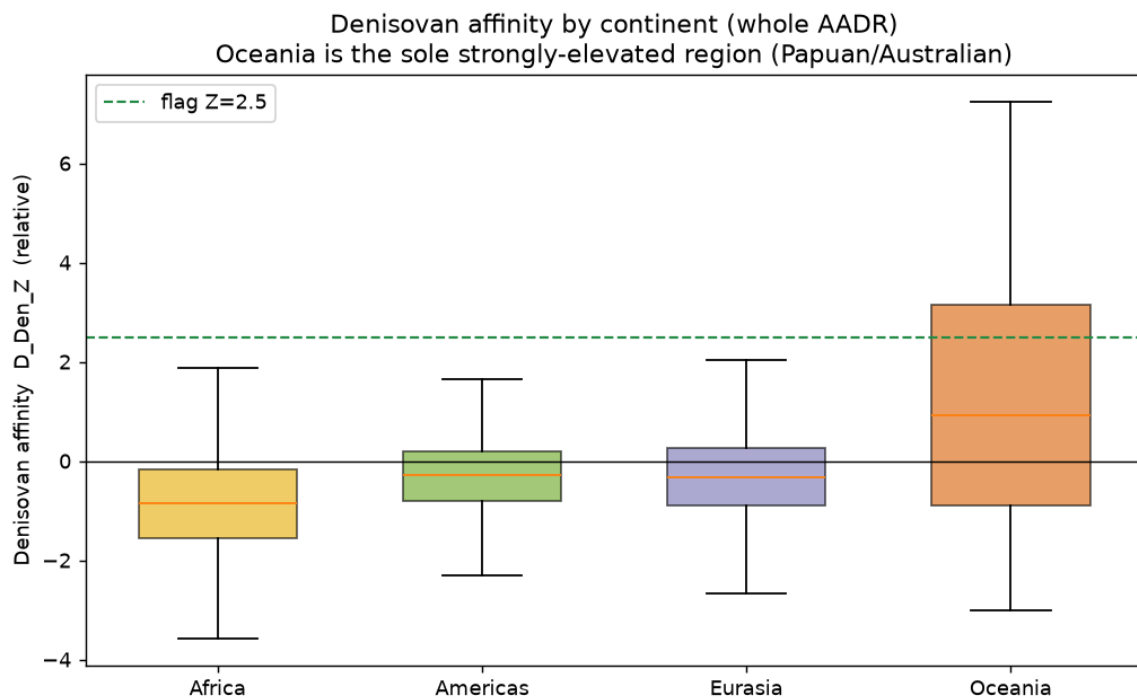


Figure 3. The archaic-ancestry plane (Neanderthal α vs Denisovan affinity), coloured by continent. Oceania occupies its own high-Denisovan corner; Neanderthal ancestry alone never clears 5%.

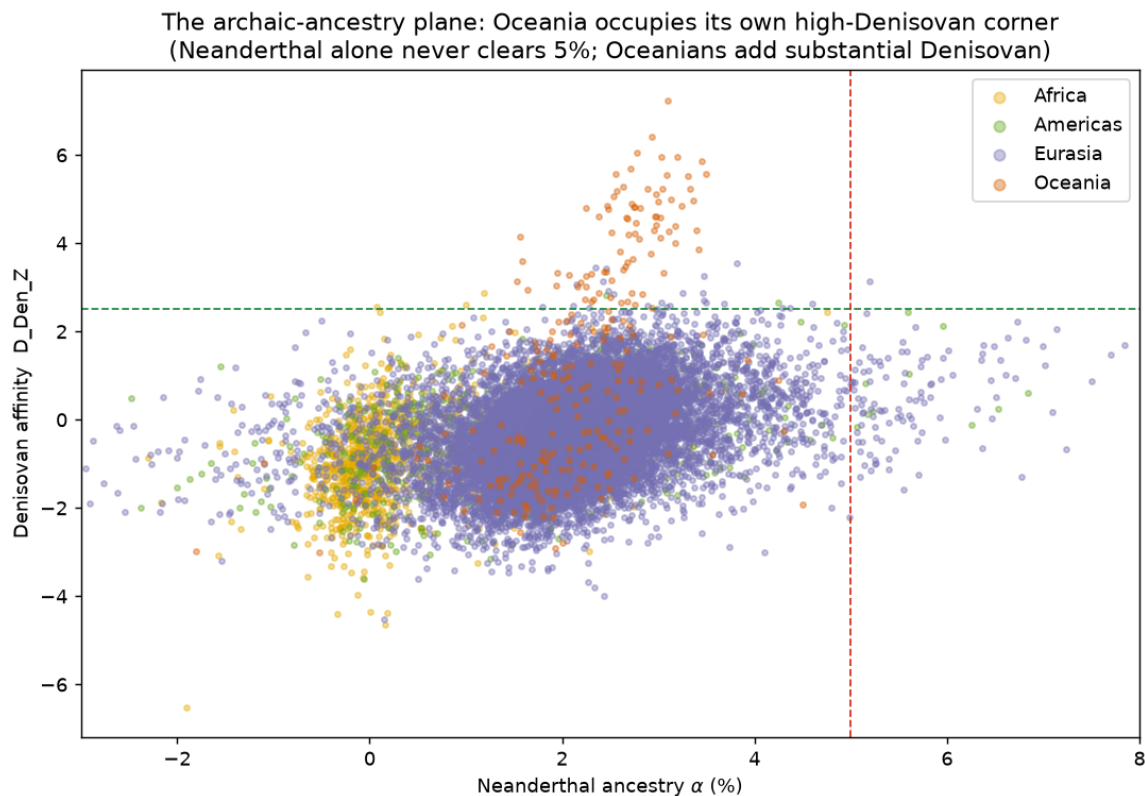
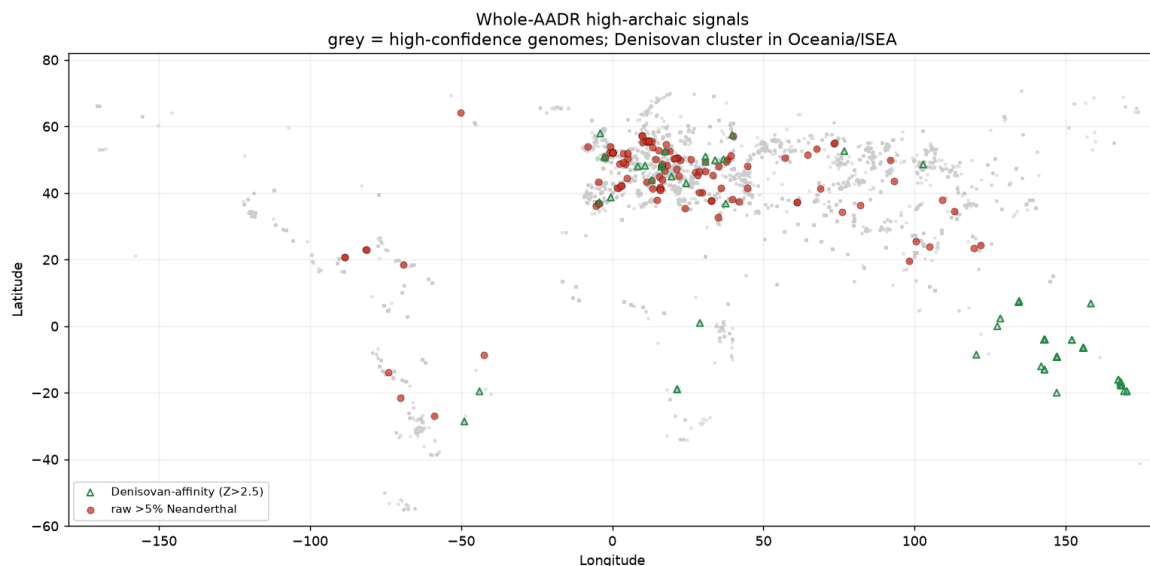


Figure 4. Whole-AADR map of high-archaic signals. Grey = high-confidence genomes; green triangles = Denisovan-affinity flags (Oceania/ISEA); red = raw >5% Neanderthal.



References

- Green R.E. et al. (2010) Science 328:710. — Reich D. et al. (2010) Nature 468:1053; (2011) AJHG 89:516. — Meyer M. et al. (2012) Science 338:222. — Patterson N. et al. (2012) Genetics 192:1065. — Prüfer K. et al. (2014) Nature 505:43. — Malaspinas A.-S. et al. (2016) Nature 538:207. — Jacobs G.S. et al. (2019) Cell 177:1010. — Mallick S. et al. (2023) Scientific Data

11:182.